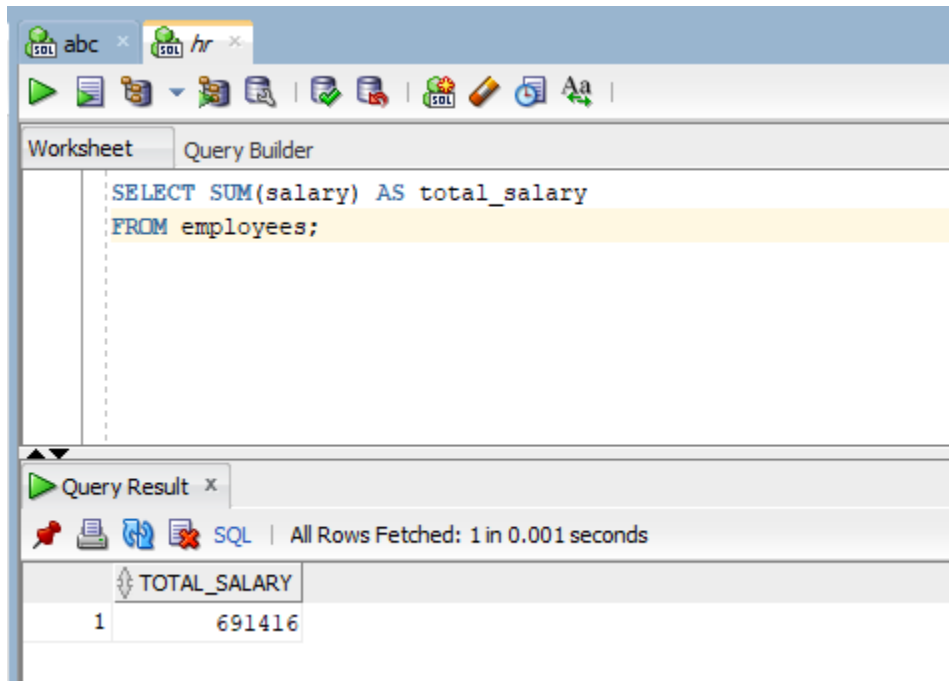


# DBS LAB 5

## IN-LAB TASKS:

### Task 1



The screenshot shows a SQL development environment with two tabs: 'abc' and 'hr'. The 'hr' tab is active, displaying a SQL query in the 'Query Builder' pane. The query is:

```
SELECT SUM(salary) AS total_salary  
FROM employees;
```

Below the query editor, the 'Query Result' pane shows the execution status: 'All Rows Fetched: 1 in 0.001 seconds'. The results are displayed in a table with one column, 'TOTAL\_SALARY', and one row with the value '691416'.

|   | TOTAL_SALARY |
|---|--------------|
| 1 | 691416       |

## Task 2

[illegible]

### Task 3

The screenshot shows a SQL query builder interface. The top toolbar includes icons for running queries, saving, and editing. The 'Query Builder' tab is active, displaying the following SQL query:

```
SELECT department_id, COUNT(*) AS total_employees
FROM employees
GROUP BY department_id;
```

Below the query editor, the 'Query Result' tab is active, showing the results of the query. The status bar indicates 'All Rows Fetched: 12 in 0.002 seconds'. The results are displayed in a table with two columns: DEPARTMENT\_ID and TOTAL\_EMPLOYEES.

|    | DEPARTMENT_ID | TOTAL_EMPLOYEES |
|----|---------------|-----------------|
| 1  | 100           | 6               |
| 2  | 30            | 6               |
| 3  | (null)        | 1               |
| 4  | 90            | 3               |
| 5  | 20            | 2               |
| 6  | 70            | 1               |
| 7  | 110           | 2               |
| 8  | 50            | 45              |
| 9  | 80            | 34              |
| 10 | 40            | 1               |
| 11 | 60            | 5               |
| 12 | 10            | 1               |

#### Task 4

The screenshot shows a SQL Query Builder window with a query entered in the Worksheet tab. The query is:

```
SELECT job_id,  
       MIN(salary) AS min_salary,  
       MAX(salary) AS max_salary  
FROM employees  
GROUP BY job_id;
```

The Query Result tab shows the results of the query. It indicates that all 19 rows were fetched in 0.002 seconds. The results are displayed in a table with three columns: JOB\_ID, MIN\_SALARY, and MAX\_SALARY.

|    | JOB_ID     | MIN_SALARY | MAX_SALARY |
|----|------------|------------|------------|
| 1  | IT_PROG    | 4200       | 9000       |
| 2  | AC_MGR     | 12008      | 12008      |
| 3  | AC_ACCOUNT | 8300       | 8300       |
| 4  | ST_MAN     | 5800       | 8200       |
| 5  | PU_MAN     | 11000      | 11000      |
| 6  | AD_ASST    | 4400       | 4400       |
| 7  | AD_VP      | 17000      | 17000      |
| 8  | SH_CLERK   | 2500       | 4200       |
| 9  | FI_ACCOUNT | 6900       | 9000       |
| 10 | FI_MGR     | 12008      | 12008      |
| 11 | PU_CLERK   | 2500       | 3100       |
| 12 | SA_MAN     | 10500      | 14000      |
| 13 | MK_MAN     | 13000      | 13000      |
| 14 | PR_REP     | 10000      | 10000      |
| 15 | AD_PRES    | 24000      | 24000      |
| 16 | SA_REP     | 6100       | 11500      |
| 17 | MK_REP     | 6000       | 6000       |
| 18 | ST_CLERK   | 2100       | 3600       |
| 19 | HR_REP     | 6500       | 6500       |

## Task 5

The screenshot shows the SQL Developer interface with a query window titled 'hr'. The query is as follows:

```
SELECT COUNT(*) AS employees_above_50000
FROM employees
WHERE salary > 50000;
```

The 'Query Result' tab shows the execution results:

| EMPLOYEES_ABOVE_50000 |
|-----------------------|
| 1                     |

The status bar indicates 'All Rows Fetched: 1 in 0.004 seconds'.

## Task 6

The screenshot shows the SQL Developer interface with a query window titled 'hr'. The query is as follows:

```
SELECT department_id, SUM(salary) AS total_salary
FROM employees
GROUP BY department_id
HAVING SUM(salary) > 100000;
```

The 'Query Result' tab shows the execution results:

| DEPARTMENT_ID | TOTAL_SALARY |        |
|---------------|--------------|--------|
| 1             | 50           | 156400 |
| 2             | 80           | 304500 |

The status bar indicates 'All Rows Fetched: 2 in 0.006 seconds'.

## Task 7

The screenshot shows a SQL query builder window with two tabs: 'Worksheet' and 'Query Builder'. The 'Query Builder' tab is active, displaying the following SQL query:

```
SELECT d.department_name, COUNT(e.employee_id) AS total_employees
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
GROUP BY d.department_name;
```

Below the query editor, the 'Query Result' tab is active, showing the results of the query. The results are displayed in a table with two columns: 'DEPARTMENT\_NAME' and 'TOTAL\_EMPLOYEES'. The table contains 11 rows of data, numbered 1 through 11.

|    | DEPARTMENT_NAME  | TOTAL_EMPLOYEES |
|----|------------------|-----------------|
| 1  | Administration   | 1               |
| 2  | Accounting       | 2               |
| 3  | Purchasing       | 6               |
| 4  | Human Resources  | 1               |
| 5  | IT               | 5               |
| 6  | Public Relations | 1               |
| 7  | Executive        | 3               |
| 8  | Shipping         | 45              |
| 9  | Sales            | 34              |
| 10 | Finance          | 6               |
| 11 | Marketing        | 2               |

The screenshot shows the SQL Developer interface. The top toolbar includes icons for running queries, saving, and editing. The 'Query Builder' tab is active, displaying the following SQL query:

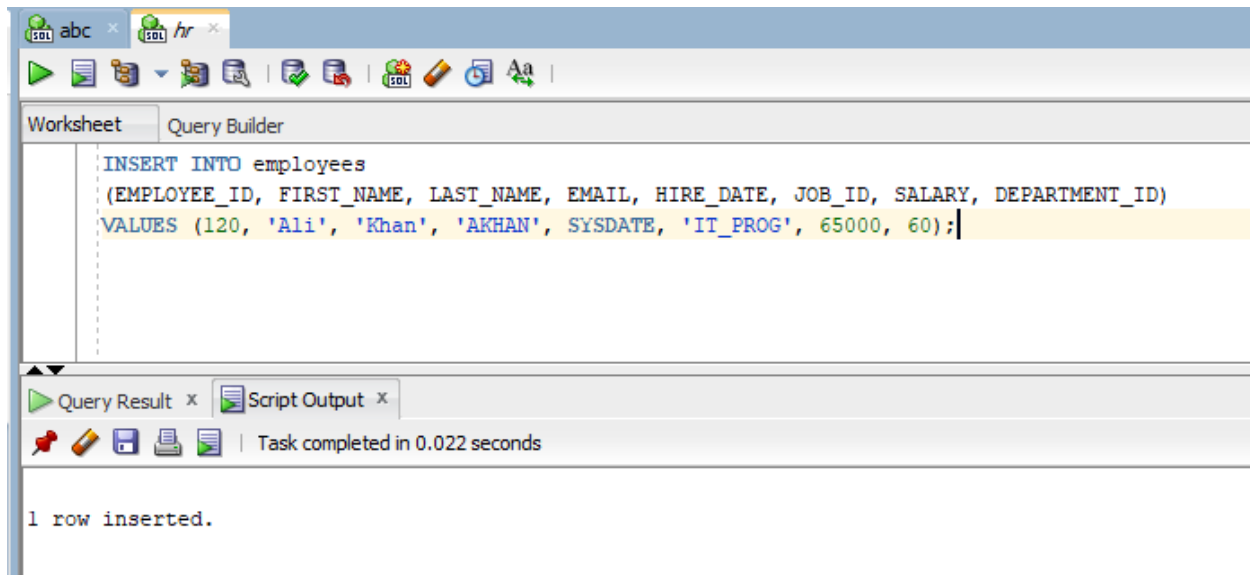
```
SELECT department_id, SUM(salary) AS total_salary
FROM employees
GROUP BY department_id
HAVING SUM(salary) > 150000;
```

Below the query editor, the 'Query Result' tab shows the results of the query. It indicates that all rows were fetched (2 rows) in 0.001 seconds. The results are displayed in a table with two columns: DEPARTMENT\_ID and TOTAL\_SALARY.

|   | DEPARTMENT_ID | TOTAL_SALARY |
|---|---------------|--------------|
| 1 | 50            | 156400       |
| 2 | 80            | 304500       |

[illegible]

## Task 10



The screenshot shows the SQL Developer interface with the 'Query Builder' tab active. The SQL statement being executed is an INSERT into the 'employees' table. The statement is: `INSERT INTO employees (EMPLOYEE_ID, FIRST_NAME, LAST_NAME, EMAIL, HIRE_DATE, JOB_ID, SALARY, DEPARTMENT_ID) VALUES (120, 'Ali', 'Khan', 'AKHAN', SYSDATE, 'IT_PROG', 65000, 60);`. The 'Script Output' tab shows the message 'Task completed in 0.022 seconds'. The 'Query Result' tab shows '1 row inserted.'

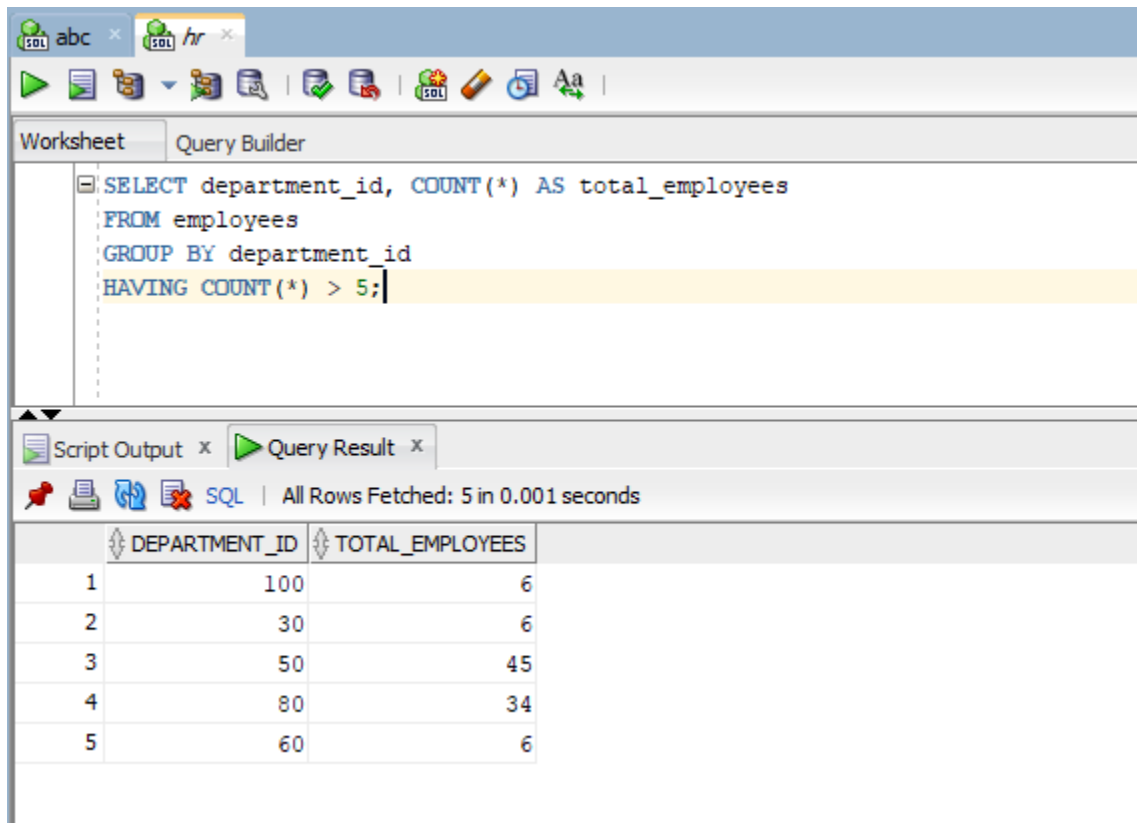
```
INSERT INTO employees
(EMPLOYEE_ID, FIRST_NAME, LAST_NAME, EMAIL, HIRE_DATE, JOB_ID, SALARY, DEPARTMENT_ID)
VALUES (120, 'Ali', 'Khan', 'AKHAN', SYSDATE, 'IT_PROG', 65000, 60);
```

Task completed in 0.022 seconds

1 row inserted.

## POST-LAB TASKS:

### Task 11



The screenshot shows the SQL Developer interface with the 'Query Builder' tab active. The SQL statement being executed is a SELECT query with a GROUP BY clause. The statement is: `SELECT department_id, COUNT(*) AS total_employees FROM employees GROUP BY department_id HAVING COUNT(*) > 5;`. The 'Script Output' tab shows the message 'All Rows Fetched: 5 in 0.001 seconds'. The 'Query Result' tab displays a table with 5 rows and 2 columns: DEPARTMENT\_ID and TOTAL\_EMPLOYEES.

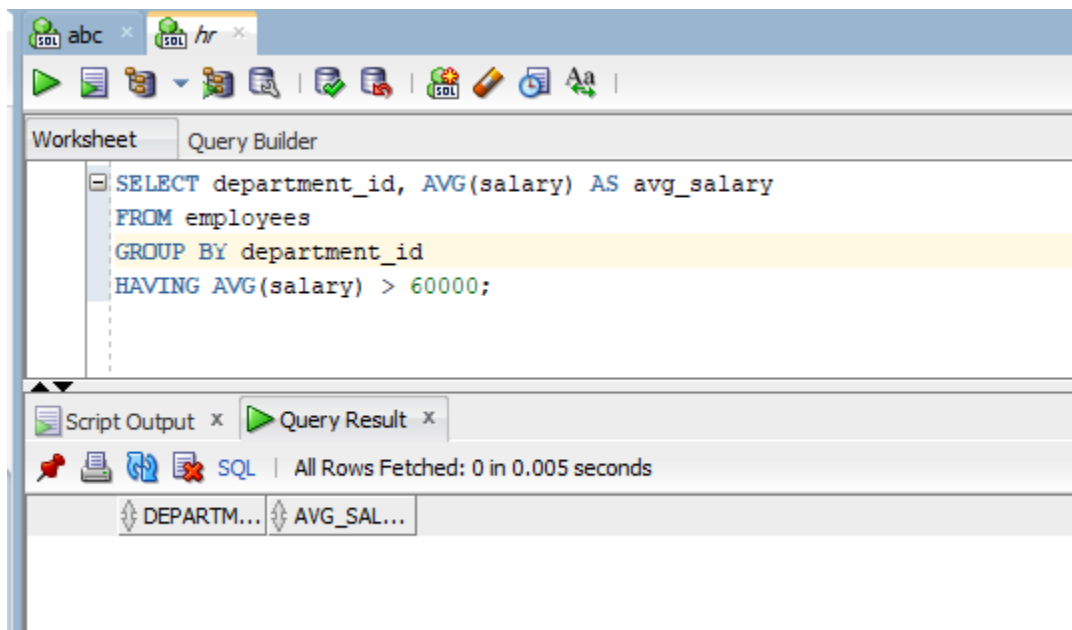
```
SELECT department_id, COUNT(*) AS total_employees
FROM employees
GROUP BY department_id
HAVING COUNT(*) > 5;
```

All Rows Fetched: 5 in 0.001 seconds

|   | DEPARTMENT_ID | TOTAL_EMPLOYEES |
|---|---------------|-----------------|
| 1 | 100           | 6               |
| 2 | 30            | 6               |
| 3 | 50            | 45              |
| 4 | 80            | 34              |
| 5 | 60            | 6               |



## Task 12

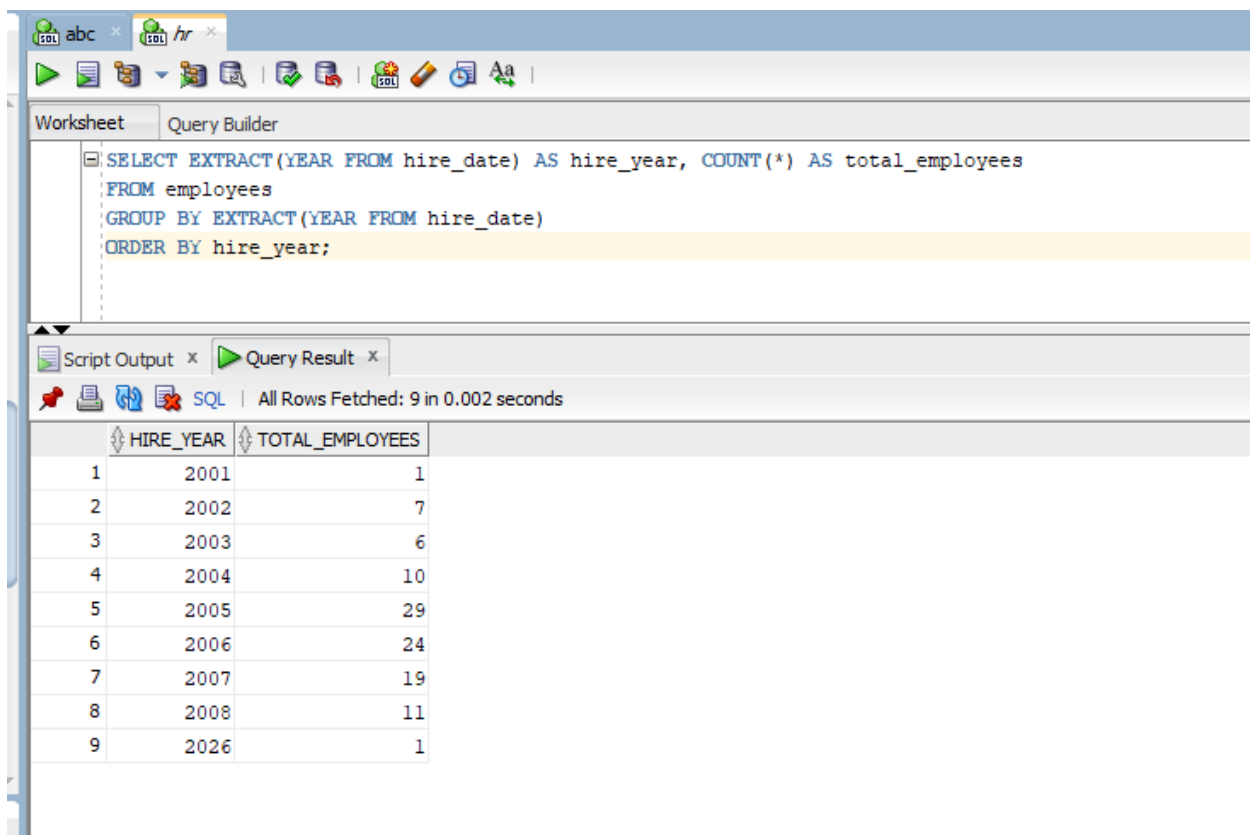


The screenshot shows the SQL Developer interface with the 'Query Builder' tab selected. The query editor contains the following SQL statement:

```
SELECT department_id, AVG(salary) AS avg_salary
FROM employees
GROUP BY department_id
HAVING AVG(salary) > 60000;
```

Below the query editor, the 'Script Output' and 'Query Result' tabs are visible. The 'Query Result' tab shows the status: 'All Rows Fetched: 0 in 0.005 seconds'. The column headers for the result are 'DEPARTM...' and 'AVG\_SAL...'.

## Task 13



The screenshot shows the SQL Developer interface with the 'Query Builder' tab selected. The query editor contains the following SQL statement:

```
SELECT EXTRACT(YEAR FROM hire_date) AS hire_year, COUNT(*) AS total_employees
FROM employees
GROUP BY EXTRACT(YEAR FROM hire_date)
ORDER BY hire_year;
```

Below the query editor, the 'Script Output' and 'Query Result' tabs are visible. The 'Query Result' tab shows the status: 'All Rows Fetched: 9 in 0.002 seconds'. The column headers for the result are 'HIRE\_YEAR' and 'TOTAL\_EMPLOYEES'.

|   | HIRE_YEAR | TOTAL_EMPLOYEES |
|---|-----------|-----------------|
| 1 | 2001      | 1               |
| 2 | 2002      | 7               |
| 3 | 2003      | 6               |
| 4 | 2004      | 10              |
| 5 | 2005      | 29              |
| 6 | 2006      | 24              |
| 7 | 2007      | 19              |
| 8 | 2008      | 11              |
| 9 | 2026      | 1               |

## Task 14

The screenshot shows a SQL IDE interface with two tabs: 'abc' and 'hr'. The 'hr' tab is active, displaying a SQL query in the 'Query Builder' pane. The query is as follows:

```
SELECT d.department_name, COUNT(e.employee_id) AS total_employees
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
GROUP BY d.department_name
ORDER BY d.department_name;
```

Below the query pane, the 'Script Output' and 'Query Result' tabs are visible. The 'Query Result' tab shows the results of the query, indicating that all 11 rows were fetched in 0.008 seconds. The results are displayed in a table with two columns: 'DEPARTMENT\_NAME' and 'TOTAL\_EMPLOYEES'.

|    | DEPARTMENT_NAME  | TOTAL_EMPLOYEES |
|----|------------------|-----------------|
| 1  | Accounting       | 2               |
| 2  | Administration   | 1               |
| 3  | Executive        | 3               |
| 4  | Finance          | 6               |
| 5  | Human Resources  | 1               |
| 6  | IT               | 6               |
| 7  | Marketing        | 2               |
| 8  | Public Relations | 1               |
| 9  | Purchasing       | 6               |
| 10 | Sales            | 34              |
| 11 | Shipping         | 45              |

## Task 15

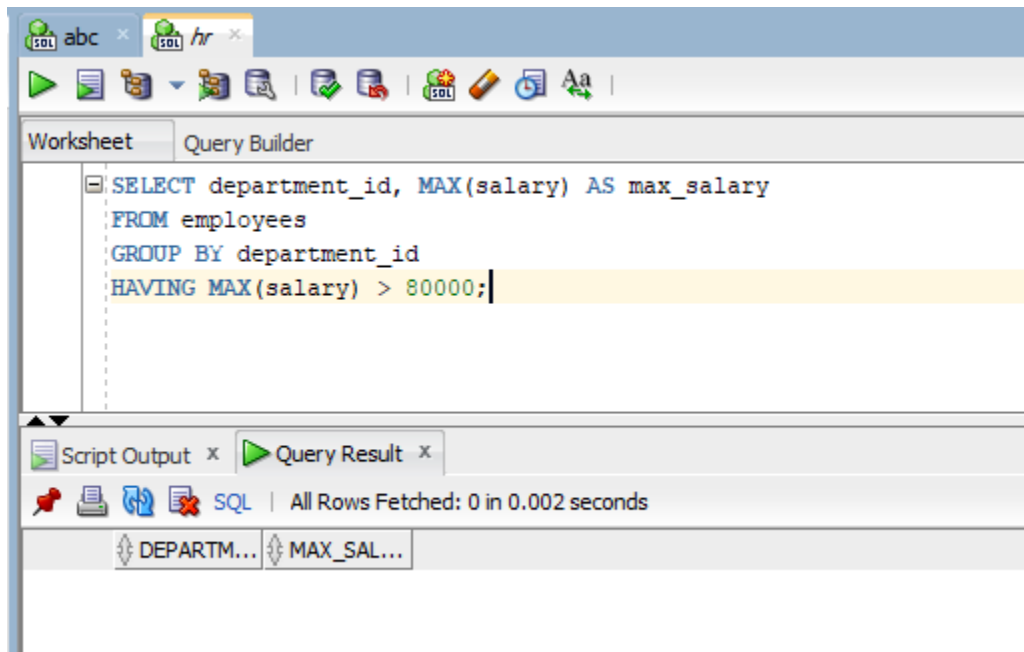
The screenshot shows a SQL IDE interface. At the top, there are two tabs: 'abc' and 'hr'. Below the tabs is a toolbar with various icons. The main window is divided into two sections: 'Worksheet' and 'Query Builder'. The 'Query Builder' section contains the following SQL query:

```
SELECT job_id, COUNT(*) AS total_employees
FROM employees
GROUP BY job_id
HAVING COUNT(*) > 3;
```

Below the query editor, there are two tabs: 'Script Output' and 'Query Result'. The 'Query Result' tab is active, showing the results of the query. The status bar indicates 'All Rows Fetched: 8 in 0.001 seconds'.

| JOB_ID       | TOTAL_EMPLOYEES |
|--------------|-----------------|
| 1 IT_PROG    | 6               |
| 2 ST_MAN     | 5               |
| 3 SH_CLERK   | 20              |
| 4 FI_ACCOUNT | 5               |
| 5 PU_CLERK   | 5               |
| 6 SA_MAN     | 5               |
| 7 SA_REP     | 30              |
| 8 ST_CLERK   | 20              |

## Task 16

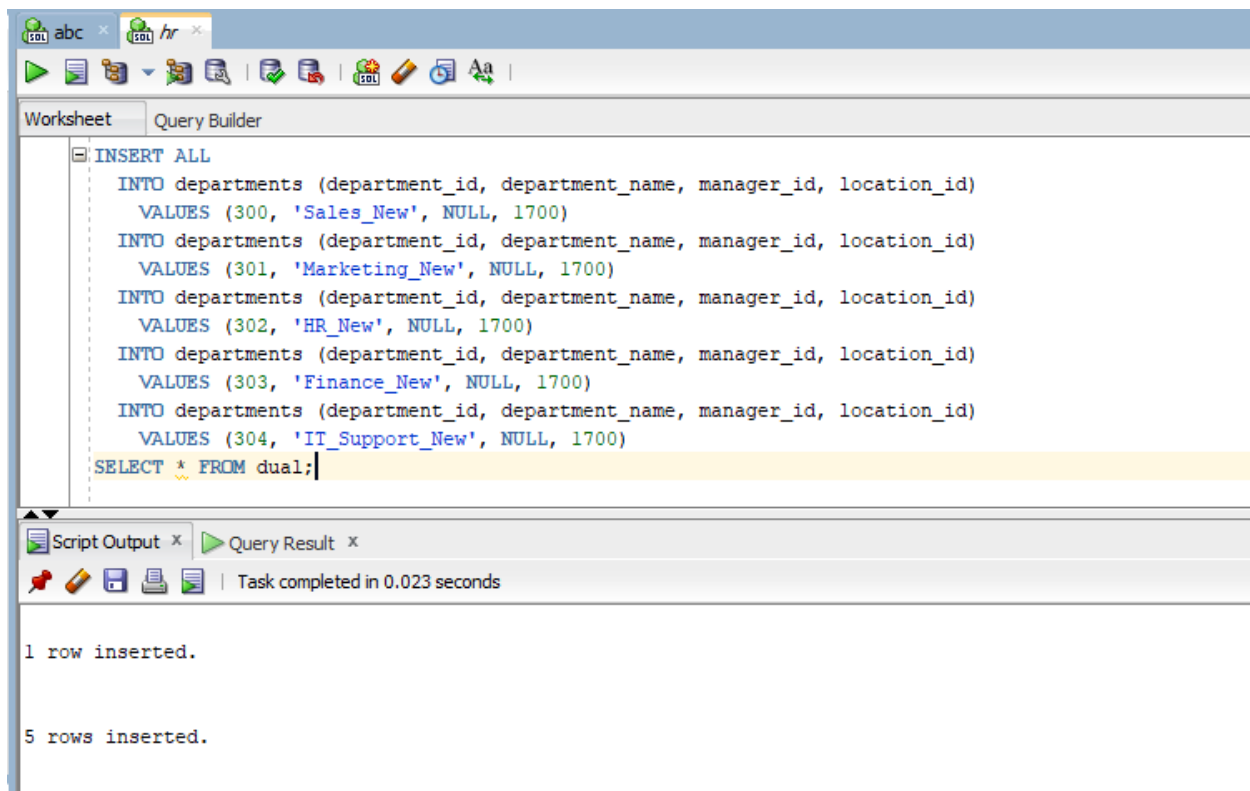


The screenshot shows the SQL Developer interface with the 'Query Builder' tab selected. The query editor contains the following SQL statement:

```
SELECT department_id, MAX(salary) AS max_salary
FROM employees
GROUP BY department_id
HAVING MAX(salary) > 80000;
```

Below the query editor, the 'Query Result' tab is active, displaying the status: 'All Rows Fetched: 0 in 0.002 seconds'. The column headers for the result are 'DEPARTM...' and 'MAX\_SAL...'.

## Task 17



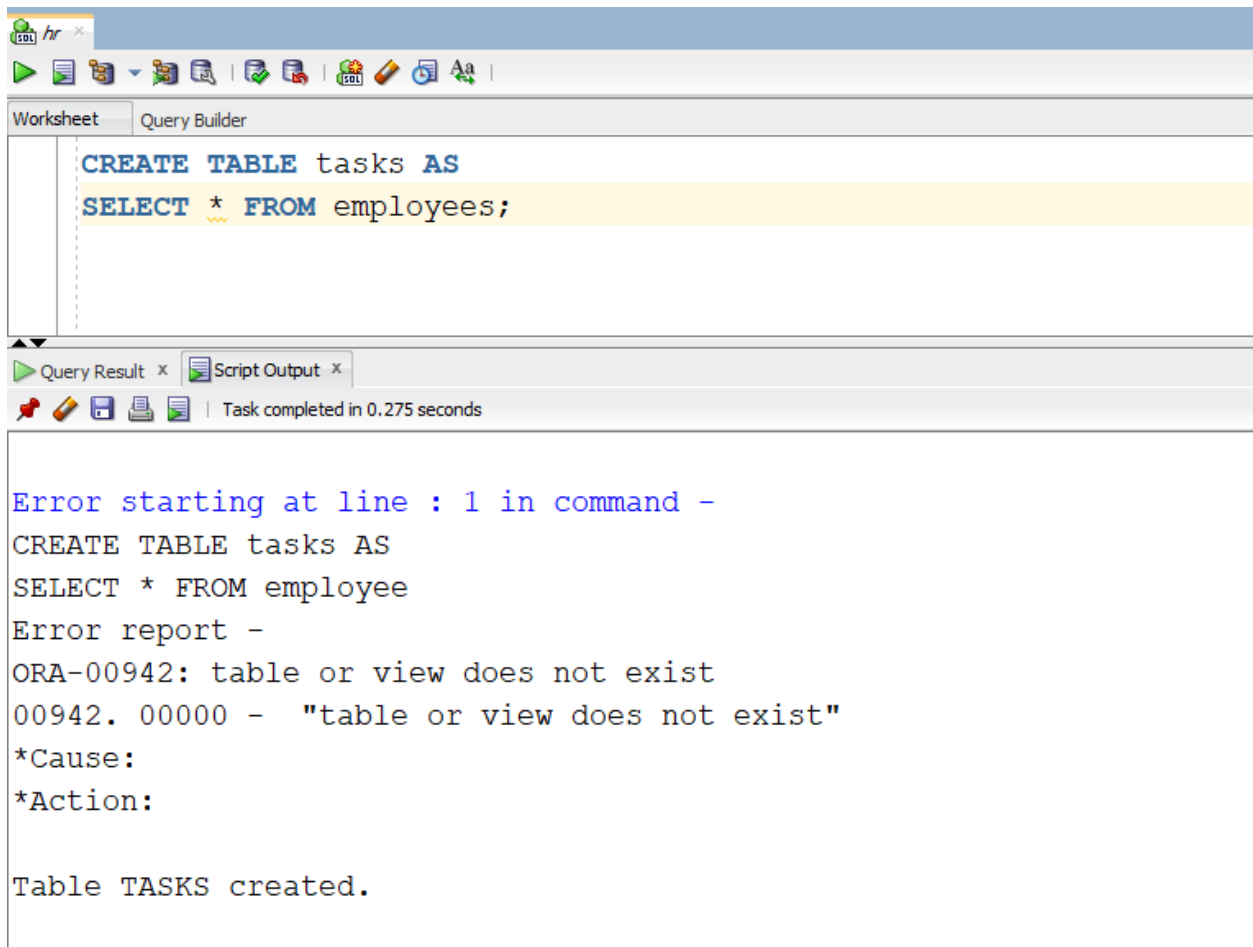
The screenshot shows the SQL Developer interface with the 'Query Builder' tab selected. The query editor contains the following SQL statement:

```
INSERT ALL
  INTO departments (department_id, department_name, manager_id, location_id)
    VALUES (300, 'Sales_New', NULL, 1700)
  INTO departments (department_id, department_name, manager_id, location_id)
    VALUES (301, 'Marketing_New', NULL, 1700)
  INTO departments (department_id, department_name, manager_id, location_id)
    VALUES (302, 'HR_New', NULL, 1700)
  INTO departments (department_id, department_name, manager_id, location_id)
    VALUES (303, 'Finance_New', NULL, 1700)
  INTO departments (department_id, department_name, manager_id, location_id)
    VALUES (304, 'IT_Support_New', NULL, 1700)
SELECT * FROM dual;
```

Below the query editor, the 'Query Result' tab is active, displaying the status: 'Task completed in 0.023 seconds'. The results are shown in a table with two rows:

| DEPARTMENT_ID | DEPARTMENT_NAME | MANAGER_ID | LOCATION_ID |
|---------------|-----------------|------------|-------------|
| 300           | Sales_New       | NULL       | 1700        |
| 301           | Marketing_New   | NULL       | 1700        |
| 302           | HR_New          | NULL       | 1700        |
| 303           | Finance_New     | NULL       | 1700        |
| 304           | IT_Support_New  | NULL       | 1700        |

## Task 18



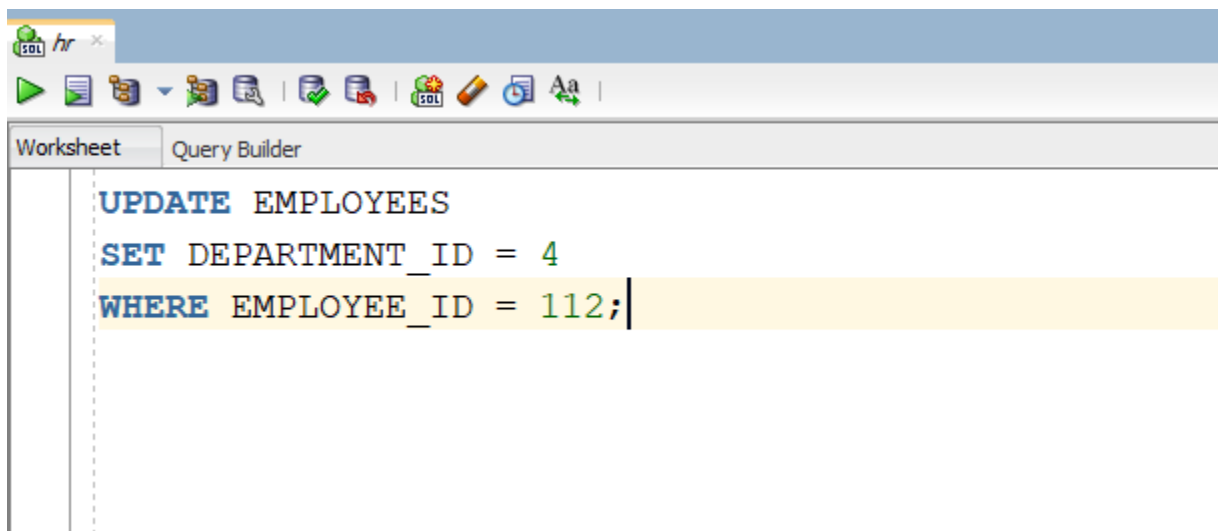
The screenshot shows the SQL Developer interface. The top toolbar includes icons for running queries, saving, and undo/redo. The 'Worksheet' tab is active, displaying the following SQL code:

```
CREATE TABLE tasks AS  
SELECT * FROM employees;
```

Below the code editor, the 'Script Output' tab is open, showing the execution results. The status bar indicates 'Task completed in 0.275 seconds'.

Error starting at line : 1 in command -  
CREATE TABLE tasks AS  
SELECT \* FROM employee  
Error report -  
ORA-00942: table or view does not exist  
00942. 00000 - "table or view does not exist"  
\*Cause:  
\*Action:  
  
Table TASKS created.

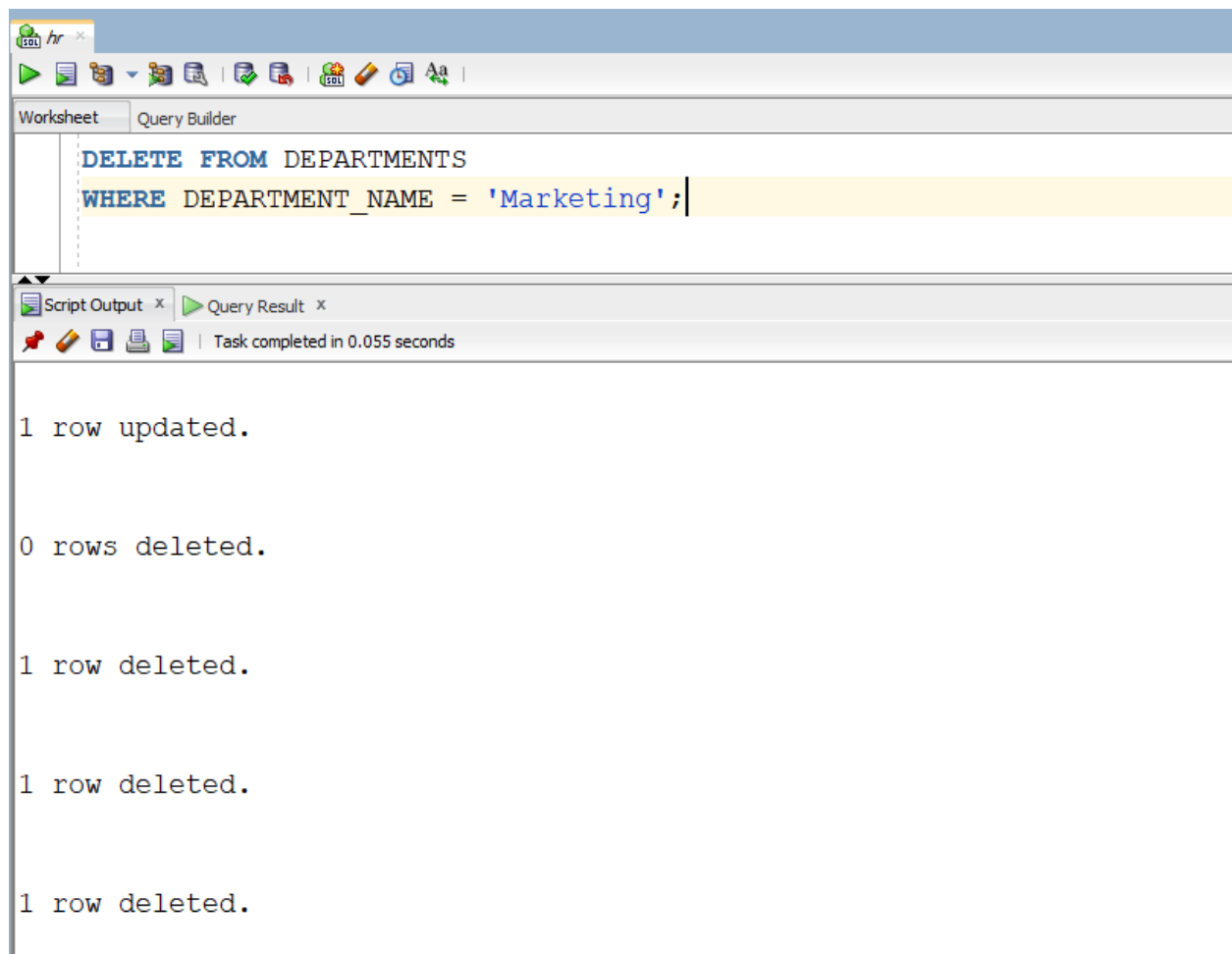
## Task 19



The screenshot shows the SQL Developer interface. The top toolbar includes icons for running queries, saving, and undo/redo. The 'Worksheet' tab is active, displaying the following SQL code:

```
UPDATE EMPLOYEES  
SET DEPARTMENT_ID = 4  
WHERE EMPLOYEE_ID = 112;
```

## Task 20



The screenshot displays a SQL query execution window. The top section, labeled 'Query Builder', contains the following SQL statement:

```
DELETE FROM DEPARTMENTS  
WHERE DEPARTMENT_NAME = 'Marketing';
```

Below the query builder, the 'Script Output' and 'Query Result' tabs are visible. The 'Query Result' tab shows the execution results:

- 1 row updated.
- 0 rows deleted.
- 1 row deleted.
- 1 row deleted.
- 1 row deleted.

The status bar at the bottom indicates 'Task completed in 0.055 seconds'.